ARASH VASHAGH

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Summary

Machine Learning Developer completing an MSc in Computer Science with a research focus on adversarial machine learning and AI security. Published researcher with hands-on experience in LLM security, machine learning systems, software development, and early-stage product development. Holds certifications from Google, Microsoft, AWS, and CompTIA. Available for full-time remote roles in Canada and the US, and on-site roles in Fredericton.

Skills

Programming: Python, Java, Kotlin, JavaScript, SQL, C++
AI/ML: PyTorch, TensorFlow, Scikit-learn, Large Language Models, LLM Agents, Agentic AI, RAG/LLMOps, Adversarial Machine Learning, Reinforcement Learning, NLP
AI Security: Adversarial Attacks, Adversarial Detection, LLM Agent Security, Robustness Evaluation, Trustworthy AI
Web, Backend & APIs: React, HTML, CSS, Streamlit, FastAPI, Flask, REST APIs, OpenAI API, JSON Processing
Cloud & Deployment: AWS, Microsoft Azure, Google Cloud, Streamlit Community Cloud, Docker
Tools & Libraries: NumPy, Pandas, Matplotlib, PIL, Jupyter, VS Code, IntelliJ IDEA, Postman, PgAdmin 4, GitHub, MLflow
Languages: Persian (Native), English (Professional)

Certifications

- **AWS:** AWS Certified AI Practitioner; AWS Certified Cloud Practitioner
- **Microsoft:** AI-900; AZ-900; DP-900; SC-900
- **Google Cloud:** Cloud Digital Leader
- **CompTIA:** Security+; Data+; Project Management Essentials; DataSys+ (results pending)
- **DeepLearning.AI:** Agentic AI

Experience

Chief Technology Officer (CTO) — Gigi Drive — Fredericton, Canada **Aug 2026 – Present**
- Part of the founding team, leading the technical direction and early-stage product development of a delivery platform.
- Responsible for technology strategy, platform architecture, system design, and technical planning for the product launch.
- Designing the backend architecture, restaurant and administrative dashboards, authentication workflows, and data models.

Graduate Research Assistant / Team Lead — UNB — Fredericton, Canada **Sep 2024 – Present**
- Conducting MSc research in adversarial machine learning, LLM security, robustness, and trustworthy AI.
- Developed ChronosAttack, showing how the arrival order of tool responses alter LLM-agent decisions.
- Co-authored CohortHijack, a robustness study of cohort-dependent single-cell annotation.
- Co-authored ConformalShift, an adversarial event-reordering attack against adaptive ECG monitoring systems.
- Co-authored a survey on adversarial attacks and a survey on adversarial defenses.
- Designed and implemented multiple defense mechanisms using Python and PyTorch (works under internal review).
- Coordinating graduate student work and managing technical and educational content for the TSAI Lab.

Graduate Teaching Assistant — UNB — Fredericton, Canada **Jan 2025 – Apr 2026**
- *Software Engineering* (Winter 2026) – Java
- *Data Communication and Network Modeling* (Fall 2025) – Python
- *Data Science for Big Data Analytics* (Winter 2025) – Python

Research Assistant — IUT — Isfahan, Iran **Feb 2021 – Feb 2024**
- Conducted research on machine learning and healthcare, resulting in two IEEE publications [1] and [2].
- Collaborated with faculty on implementation, paper writing, and presenting at conferences.

Teaching Assistant — IUT — Isfahan, Iran **Feb 2021 – Feb 2024**
- Courses: Artificial Intelligence, Embedded Systems, Data Structures, and Programming Labs in C++, Java, and Python.
- Gave tutorials and graded assignments.

Education

Master's Candidate — University of New Brunswick (UNB), Fredericton, Canada **Sep 2024 – Aug 2026**
- Awarded Graduate Scholarship and Research Grant.
- Major: Computer Science
- Research Focus: Adversarial Machine Learning

Bachelor of Science — Isfahan University of Technology (IUT), Isfahan, Iran **Sep 2019 – Feb 2024**
- Major: Computer Engineering
- Minor: Intelligent Systems

Selected Projects

ChronosAttack: Adversarial Tool Scheduling Attacks on LLM Agents **Aug 2026**

- Developed a delay-only adversarial scheduling attack against tool-using LLM agents.
- Demonstrated that changing when authentic tool responses arrive can change an agent's final decision.
- Evaluated GPT-5.6 Sol, Gemini 3.6 Flash, DeepSeek V4 Flash, and Claude Sonnet 4.6.
- Found substantial model-dependent sensitivity to observation order.
- Technologies: Python, LLM APIs, agentic AI, adversarial machine learning, AI security, robustness evaluation.
- Paper: Research Square | Source code: GitHub.

ConformalShift: Event-Reordering Attacks on Adaptive ECG Monitoring **Jun 2026 – Aug 2026**

- Developed a targeted attack against adaptive ECG monitoring systems that learn from feedback provided by doctors.
- Demonstrated that changing only the order of authentic feedback can reduce the detection of clinically important abnormal heartbeats.
- Designed and evaluated the attack using ECG datasets and multiple machine learning classifiers.
- Technologies: Python, Scikit-learn, adaptive conformal prediction, ECG analysis, healthcare AI.
- Paper: arXiv | Source code: GitHub.

PaperWise AI: Deployed Agentic Research Paper RAG / LLMOps App **Jun 2026**

- Built and deployed a public Streamlit app for research paper question answering.
- Implemented PDF ingestion, text extraction, chunking, TF-IDF retrieval, OpenAI API integration, and offline fallback.
- Applied agentic design through suggested questions, reviewer critique mode, structured paper summaries, multi-paper comparison, and PDF answer export.
- Displayed RAG pipeline metrics including retrieval scores, response latency, retrieved chunks, and citation usage.
- Technologies: Python, Streamlit, Scikit-learn, OpenAI API, pypdf, ReportLab, Streamlit Community Cloud.
- Live app: Streamlit | Source code: GitHub.

Voluntary Educator (AI Security)

AI Security and Adversarial Machine Learning (AML) Course — Online **Jun 2026 – Present**

- Creating a Persian technical course on AI Security and AML to support learners facing language barriers.
- Developed slides and scripts on machine learning foundations for AML.
- Published course videos through a structured YouTube playlist and shared course materials on GitHub.